Ex 1.3

Q1 :-

(a-) BA

BA

$(4\times5)\cdot(4\times5)$

undefined.

(b-) AB^T

$(4\times5)\cdot(5\times4)$

(4×4) defined

(c-) AC+D

$=(4\times5)(5\times2)+(4\times2)$

$=(4\times2)+(4\times2)$

$=(4\times2)$ defined

(d-) E(AC)

$=(5\times4)[(4\times5)\cdot(5\times2)]$

$=(5\times4)\cdot(4\times2)$

$=(5\times2)$ defined

(e-) $A-3E^t$

$(4\times5)-3(4\times5)$

$(4\times5)-(4\times5)$

= (4 x 5) defined.

(f-) B(5B+A)

= (4x5) [5(4x5) + (4x5)]

= (4x5) [(4x5) + (4x5)]

= (4x5).(4x5)

= ~~4x~~ undefined.

Q2:—

(a-) CD^T

= (5x2).(2x4)

= (5x4) defined.

(b-) DC

= (4x2).(5x2)

undefined

(c-) BC-3D

$= (4\times5)\cdot(5\times2) - 3(4\times2)$

$= (4\times2) - (4\times2)$

$= (4\times2)$ defined.

(d-) $D^T(BE)$

$= (2\times4)[(4\times5)\cdot(5\times4)]$

$= (2\times4)\cdot(4\times5)$

$= (2\times5)$ defined

(e-) $B^TD + ED$

$= (5\times4)\cdot(4\times2) + (5\times4)\cdot(4\times2)$

$= (5\times2) + (5\times2)$

$= (5\times2)$ defined

(f-) $BA^T + D$

$= (4\times5)(5\times4) + (4\times2)$

$= (4\times4) + (4\times2)$

$=$ undefined.

Q3:—

(a-) D+E

$$= \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 6 & 5 \\ -2 & 1 & 3 \\ 7 & 3 & 7 \end{bmatrix}$$

(b-) D-E

$$= \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

(c-) 5A

$$= 5 \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 0 \\ -5 & 10 \\ 5 & 5 \end{bmatrix}$$

(d-) $-7C$

$$= -7 \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & -28 & -14 \\ -21 & -7 & -35 \end{bmatrix}$$

(e-) $2B - C$ ~~$(2\times2) - (2\times3)$~~

$$= 2\begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -2 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

= undefined

(f-) $4E - 2D$

$$= 4\begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} - 2\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & 4 & 12 \\ -4 & 4 & 8 \\ 16 & 4 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 10 & 4 \\ -2 & 0 & 2 \\ 6 & 4 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -6 & 8 \\ -2 & 4 & 6 \\ 10 & 0 & 4 \end{bmatrix}$$

(g-) $-3(D+2E)$

$$= -3\left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 12 & 2 & 6 \\ -2 & 2 & 4 \\ 8 & 2 & 6 \end{bmatrix}\right)$$

$$= -3 \begin{bmatrix} 13 & 7 & 8 \\ -3 & 2 & 5 \\ 11 & 4 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} -39 & -21 & -24 \\ 9 & -6 & -15 \\ -33 & -12 & -30 \end{bmatrix}$$

(h-) A-A

$$= \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(i-) tr(D)

$$= tr \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$= (1) + (0) + (4)$$

$$= 5$$

(j-) tr(D-3E)

$$= tr\left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} +18 & 3 & 9 \\ -3 & 3 & 6 \\ 12 & 3 & 9 \end{bmatrix} \right)$$

$$= tr \begin{bmatrix} -17 & 2 & -7 \\ 2 & -3 & -5 \\ -9 & -1 & -5 \end{bmatrix}$$

$= (-17)+(-3)+(-5)$

$= -17-3-5$

$= -20-5$

$= -25$

(k-) 4 tr(7B)

$$= 4 \; tr \begin{bmatrix} 28 & -7 \\ 0 & 14 \end{bmatrix}$$

$= 4 \; (28+14)$

$= 4 \; (42)$

$= 168$

(l-) tr(A)

$$= \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

= Undefined.

Q4 :—

(a-) $2A^T + C$

$$= 2\begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -2 & 2 \\ 0 & 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 2 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

(b-) $D^T - E^T$

$$= \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 0 & -1 \\ 4 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

(c-) $(D-E)^T$

$$D-E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$(D-E)^T = \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}^t$$

$$= \begin{bmatrix} -5 & 0 & -1 \\ 4 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

(d-) $B^T + 5C^T$ $\quad (2\times 2) + (3\times 2)$

$$= \begin{bmatrix} 4 & 0 \\ -1 & 2 \end{bmatrix} + 5\begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix}$$

= undefined

(e-) $\frac{1}{2}C^T - \frac{1}{4}A$

$$= \frac{1}{2}\begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} - \frac{1}{4}\begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 3/2 \\ 2 & 1/2 \\ 1 & 5/2 \end{bmatrix} - \begin{bmatrix} 3/4 & 0 \\ -1/4 & 1/2 \\ 1/4 & 1/4 \end{bmatrix}$$

$$= \begin{bmatrix} -1/4 & 3/2 \\ 9/4 & 0 \\ 3/4 & 9/4 \end{bmatrix}$$

(f-) $B - B^T$

$$= \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(g-) $2E^T - 3D^T$

$$= 2\begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix} - 3\begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -2 & 8 \\ 2 & 2 & 2 \\ 6 & 4 & 6 \end{bmatrix} - \begin{bmatrix} 3 & -3 & 9 \\ 15 & 0 & 6 \\ 6 & 3 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 1 & -1 \\ -13 & 2 & -4 \\ 0 & 1 & -6 \end{bmatrix}.$$

(h-) $(2E^T - 3D^T)^T$

$$(2E^T - 3D^T)^T = \begin{bmatrix} 9 & 1 & -1 \\ -13 & 2 & -4 \\ 0 & 1 & -6 \end{bmatrix}^T$$

$$= \begin{bmatrix} 9 & -13 & 0 \\ 1 & 2 & 1 \\ -1 & -4 & -6 \end{bmatrix}$$

(i-) (CD)E

$$= \left(\begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \right) E$$

3×3

$$= \begin{bmatrix} (1)(1)+(4)(-1)+(2)(3) & (1)(5)+(4)(0)+(2)(2) & (1)(2)+(4)(1)+(2) \\ (3)(1)+(1)(-1)+(5)(3) & (3)(5)+(1)(0)+(5)(2) & (3)(2)+(1)(1)+(5 \\ \end{bmatrix}$$

$$= \begin{bmatrix} 1-4+6 & 5+0+4 & 2+4+8 \\ 3-1+15 & 15+0+10 & 6+1+20 \end{bmatrix} E$$

$$= \begin{bmatrix} 3 & 9 & 14 \\ 17 & 25 & 27 \end{bmatrix} \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} (3)(6)+(9)(-1)+(14)(4) & (3)(1)+(9)(1)+(14)(1) & (3)(3)+(9)(2)+(14)(\\ (17)(6)+(25)(-1)+(27)(4) & (17)(1)+(25)(1)+(27)(1) & (17)(3)+(25)2+27(\end{bmatrix}$$

$$= \begin{bmatrix} 18-9+56 & 3+9+14 & 9+18+42 \\ 102-25+108 & 17+25+27 & 51+50+81 \end{bmatrix}$$

$$= \begin{bmatrix} 65 & 26 & 69 \\ 185 & 69 & 182 \end{bmatrix}$$

(j-) C(BA)

$$= \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \left(\begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} \text{undefined} \end{bmatrix}$$

undefined.

(k-) tr(DE^T)

$$= tr\left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}\begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix}\right)$$

$$= tr\begin{bmatrix} 6+5+6 & -1+5+4 & 4+5+6 \\ -6+0+3 & 1+0+2 & -4+0+3 \\ 18+2+12 & -3+2+8 & 12+2+12 \end{bmatrix}$$

$$= tr\begin{bmatrix} 17 & 8 & 15 \\ -3 & 3 & -1 \\ 32 & 7 & 26 \end{bmatrix}$$

$$= (17)+(3)+26$$

$$= 46$$

(ب-) tr (BC)

$$= tr\left(\begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}\right)$$

$$= tr\begin{bmatrix} 4-3 & 16-1 & 8-5 \\ 0+6 & 0+2 & 0+10 \end{bmatrix}$$

$$= tr\begin{bmatrix} 1 & 15 & 3 \\ 6 & 2 & 10 \end{bmatrix}$$

undefined.

Q5 and Q6 are same.

Q7:-

(a-) the first row of AB

$$= \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 18-0+49 & -6-2+49 & 12-6+35 \\ 36+0+28 & -12+5+28 & 24+15+20 \\ 0+0+63 & 0+4+63 & 0+12+45 \end{bmatrix}$$

$$= \begin{bmatrix} 67 & 41 & 41 \\ 64 & 16 & 59 \\ 63 & 67 & 57 \end{bmatrix}$$

⇒ the first row of AB is [67 41 41]

Q 7, 8 are same.

Exercise 1.4

Question : 1

(a-)

Associative law for matrix addition

$A+(B+C)=(A+B)+C$

L.H.S

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ -2 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 2 \\ 0 & -2 \end{bmatrix}$$

R.H.S

$$= \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \right) + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 2 \\ 0 & -2 \end{bmatrix}$$

L·H·S = R·H·S

(b-) A·P for multiplication

$A(BC) = (AB)C$

L·H·S

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -6 & -4 \\ 16 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} -34 & -21 \\ 52 & 28 \end{bmatrix}$$

R·H·S

$$= \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \right) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 10 \\ 4 & -12 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -34 & -21 \\ 52 & 28 \end{bmatrix}$$

L·H·S = R·H·S

(c-) left distributive law

$A(B+C) = AB + AC$

L·H·S

A.

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ -2 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 15 \\ 0 & -18 \end{bmatrix}$$

R.H.S

$$= \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \right) + \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} -1 & 10 \\ 4 & -12 \end{bmatrix} + \begin{bmatrix} 15 & 5 \\ -4 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 15 \\ 0 & -18 \end{bmatrix}$$

L.H.S = R.H.S

(d-) $(a+b)C = aC + bC$

$$(4-7) \begin{bmatrix} 4 & 1 \\ 3 & -2 \end{bmatrix} = 4 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} + (-7) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$-3 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} 16 & 4 \\ -12 & -8 \end{bmatrix} + \begin{bmatrix} -28 & -7 \\ 21 & 14 \end{bmatrix}$$

$$\begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix} = \begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix}$$

L.H.S = R.H.S.

Q2:-

(a-) $a(BC) = (aB)C = B(aC)$

$$4\left(\begin{bmatrix}0 & 2\\ 1 & -4\end{bmatrix}\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}\right) = \left(4\begin{bmatrix}0 & 2\\ 1 & -4\end{bmatrix}\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}\right) = \begin{bmatrix}0 & 2\\ 1 & -4\end{bmatrix}4\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}$$

$$4\begin{bmatrix}-6 & -4\\ 16 & 9\end{bmatrix} = \begin{bmatrix}0 & 8\\ 4 & -16\end{bmatrix}\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix} = \begin{bmatrix}0 & 2\\ 1 & -4\end{bmatrix}\begin{bmatrix}16 & 4\\ -12 & -8\end{bmatrix}$$

$$\begin{bmatrix}-24 & -16\\ 64 & 36\end{bmatrix} = \begin{bmatrix}-24 & -16\\ 64 & 36\end{bmatrix} = \begin{bmatrix}-24 & -16\\ 64 & 36\end{bmatrix}$$

proved

(b-) $A(B-C) = AB - AC$

$$\begin{bmatrix}3 & -1\\ 2 & 4\end{bmatrix}\left(\begin{bmatrix}0 & 2\\ 1 & 4\end{bmatrix} - \begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}\right) = \begin{bmatrix}3 & -1\\ 2 & 4\end{bmatrix}\begin{bmatrix}0 & 2\\ 1 & 4\end{bmatrix} - \begin{bmatrix}3 & -1\\ 2 & 4\end{bmatrix}\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}$$

$$\begin{bmatrix}3 & -1\\ 2 & 4\end{bmatrix}\begin{bmatrix}-4 & 1\\ 4 & -2\end{bmatrix} = \begin{bmatrix}-1 & 10\\ 4 & -12\end{bmatrix} - \begin{bmatrix}15 & 5\\ -4 & -6\end{bmatrix}$$

$$\begin{bmatrix}-16 & 5\\ 8 & -6\end{bmatrix} = \begin{bmatrix}-16 & 5\\ 8 & -6\end{bmatrix}$$

proved.

(c) $(B+C)A = BA + CA$

$$\left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}\right)\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ -2 & -6 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ -5 & -17 \end{bmatrix} + \begin{bmatrix} 14 & 10 \\ -13 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 18 & 8 \\ -18 & -22 \end{bmatrix} = \begin{bmatrix} 18 & 8 \\ -18 & -22 \end{bmatrix}$$

proved.

(d) $a(bC) = (ab)C$

$$4\left(-7\begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}\right) = [(4)(-7)]\begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$4\begin{bmatrix} -28 & -7 \\ 21 & 14 \end{bmatrix} = -28\begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} -112 & -28 \\ 84 & 56 \end{bmatrix} = \begin{bmatrix} -112 & -28 \\ 84 & 56 \end{bmatrix}$$

proved :

Q3:—

(a) $(A^t)^t = A$

$$\left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}^t\right)^t = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}^T = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

(b)

$$(AB)^T = B^T A^T$$

$$\left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}\right)^T = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}^T \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}^T$$

$$\begin{bmatrix} -1 & 10 \\ 4 & -12 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 10 & -12 \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 10 & -12 \end{bmatrix}$$

Q4:—

(a) $(A+B)^T = A^T + B^T$

$$= \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}\right)^T = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}^T + \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}^T$$

$$= \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix}^T = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 3 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 1 & 0 \end{bmatrix}$$

(b) $(aC)^T = a(C)^T$

$$\left(4\begin{bmatrix}4 & 1\\ -3 & -2\end{bmatrix}\right)^T = 4\begin{bmatrix}4 & 1\\ 3 & -2\end{bmatrix}^T$$

$$\begin{bmatrix}16 & 4\\ -12 & -8\end{bmatrix}^T = 4\begin{bmatrix}4 & -3\\ 1 & -2\end{bmatrix}$$

$$\begin{bmatrix}16 & -12\\ 4 & -8\end{bmatrix} = \begin{bmatrix}16 & -12\\ 4 & -8\end{bmatrix}$$

Q5:—

$$A = \begin{bmatrix}2 & -3\\ 4 & 4\end{bmatrix}$$

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$ ← from 1.4.5

$$\left(\frac{1}{|A|}(\text{adj } A)\right)$$

$$= \frac{1}{(2)(4)-(-3)(4)}\begin{bmatrix}4 & -(-3)\\ -(4) & 2\end{bmatrix}$$

$$= \frac{1}{20}\begin{bmatrix}4 & 3\\ -4 & 2\end{bmatrix}$$

$$\begin{bmatrix}1/5 & 3/20\\ -1/5 & 1/10\end{bmatrix}$$

Q6:—

$$B = \begin{bmatrix}3 & 1\\ 5 & 2\end{bmatrix}$$

$$B^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{(3)(2)-(1)(5)}\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

$$= 1\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

Q7:—

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$C = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{(2)(3)-(0)(0)}\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \frac{1}{6}\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}$$

Q8:—

$$D = \begin{bmatrix} 6 & 4 \\ -2 & -1 \end{bmatrix}$$

$$D^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{(6)(-1)-(4)(-2)}\begin{bmatrix} -1 & -4 \\ 2 & 6 \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} -1 & -4 \\ 2 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & -2 \\ 1 & 3 \end{bmatrix}$$

Q9:—

let $A = \begin{bmatrix} \frac{1}{2}(e^{x}+e^{-x}) & \frac{1}{2}(e^{x}-e^{-x}) \\ \frac{1}{2}(e^{x}-e^{-x}) & \frac{1}{2}(e^{x}+e^{-x}) \end{bmatrix}$

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{\left[\frac{1}{2}(e^{x}+e^{-x})\right]^2 - \left[\frac{1}{2}(e^{x}-e^{-x})\right]^2}\begin{bmatrix} \frac{1}{2}(e^{x}+e^{-x}) & -\frac{1}{2}(e^{x}-e^{-x}) \\ -\frac{1}{2}(e^{x}-e^{-x}) & \frac{1}{2}(e^{x}+e^{-x}) \end{bmatrix}$$

$$\frac{1}{\frac{e^{2x}+e^{-2x}+2}{4}-\left(\frac{e^{-2x}+e^{-2x}-2}{4}\right)}\begin{bmatrix}\frac{1}{2}(e^x+e^{-x}) & -\frac{1}{2}(e^x-e^{-x})\\ -\frac{1}{2}(e^x-e^{-x}) & \frac{1}{2}(e^x+e^{-x})\end{bmatrix}$$

$$=\frac{1}{\frac{e^{2x}+e^{-2x}+2-e^{2x}+2-e^{-2x}}{4}}\begin{bmatrix}\frac{1}{2}(e^x+e^{-x}) & -\frac{1}{2}(e^x-e^{-x})\\ -\frac{1}{2}(e^x-e^{-x}) & \frac{1}{2}(e^x+e^{-x})\end{bmatrix}$$

$$=\frac{4}{4}\begin{bmatrix}\frac{1}{2}(e^x+e^{-x}) & -\frac{1}{2}(e^x-e^{-x})\\ -\frac{1}{2}(e^x-e^{-x}) & \frac{1}{2}(e^x+e^{-x})\end{bmatrix}$$

$$=\begin{bmatrix}\frac{1}{2}(e^x+e^{-x}) & -\frac{1}{2}(e^x-e^{-x})\\ -\frac{1}{2}(e^x-e^{-x}) & \frac{1}{2}(e^x+e^{-x})\end{bmatrix}\text{Ans}$$

Q 10:—

let $A\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$

$$A^{-1}=\frac{1}{ad-bc}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$= \frac{1}{(\cos\theta)(\cos\theta)-(-\sin\theta)(\sin\theta)} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$= \frac{1}{\cos^2\theta + \sin^2\theta} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$= 1 \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \text{Ans}$$

Q11:—

$$(A^T)^T = (A^T)^{-1}$$

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix} \qquad A^T = \begin{bmatrix} 2 & 4 \\ -3 & 4 \end{bmatrix}$$

$$(A^{-1})^T = \left(\frac{1}{|A|} \text{Adj}\, A\right)^T \Rightarrow (A^T)^{-1} = \frac{1}{|A^T|} \text{Adj}\, A^T$$

$$= \left(\frac{1}{20}\begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix}\right)^T, \qquad = \frac{1}{20}\begin{bmatrix} 4 & -4 \\ 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1/5 & +3/20 \\ -4/5 & 1/10 \end{bmatrix}^T, \qquad = \begin{bmatrix} 1/5 & -1/5 \\ 3/10 & 1/10 \end{bmatrix}$$

$$= \begin{bmatrix} 1/5 & -1/5 \\ 3/10 & 1/10 \end{bmatrix}$$

L.H.S = R.H.S

Hence, proved.

Q 12:—

$(A^{-1})^{-1} = A$

$$A^{-1} = \begin{bmatrix} 1/5 & 3/20 \\ -1/5 & 1/10 \end{bmatrix}$$

$$|A^{-1}| = \frac{1}{50} + \frac{3}{100}$$

$$= 2+3/100$$

$$= 1/10$$

$$\text{Adj } A^{-1} = \begin{bmatrix} 1/10 & -3/20 \\ 1/5 & 1/5 \end{bmatrix}$$

$$(A^{-1})^{-1} = \frac{1}{|A^{-1}|} \text{Adj } A^{-1}$$

$$= 20 \begin{bmatrix} 1/10 & -3/20 \\ 2/5 & 1/5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$$

Q 13:

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

L.H.S $= (ABC)^{-1}$

$$= \left(\begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} -9 & -4 \\ 32 & 12 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} -18 & -12 \\ 64 & 36 \end{bmatrix}^{-1}$$

$$= \frac{1}{(18)(36)-(-12)(64)} \begin{bmatrix} 36 & 12 \\ -64 & -18 \end{bmatrix}$$

$$= \frac{1}{120} \begin{bmatrix} 36 & 12 \\ -64 & -18 \end{bmatrix}$$

$$= \begin{bmatrix} 3/10 & 1/10 \\ -8/15 & -3/20 \end{bmatrix}$$

$$R.H.S = C^{-1}B^{-1}A^{-1}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 1/5 & 3/20 \\ -1/5 & 1/10 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1/2 \\ -5/3 & 1 \end{bmatrix} \begin{bmatrix} 1/5 & 3/20 \\ -1/5 & 1/10 \end{bmatrix}$$

$$= \begin{bmatrix} 3/10 & 1/10 \\ -8/15 & -3/20 \end{bmatrix}$$

$L.H.S = R.H.S$

Hence proved.

Q14:—

$$(ABC)^T = C^T B^T A^T$$

$$L.H.S = (ABC)^T$$

$$= \begin{bmatrix} -18 & 64 \\ -12 & 36 \end{bmatrix}^T$$

$$= \begin{bmatrix} -18 & 64 \\ -12 & 36 \end{bmatrix}$$

$$R.H.S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}^T \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}^T \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}^T$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 10 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -18 & 64 \\ -12 & 36 \end{bmatrix}$$

L.H.S = R.H.S

Hence, proved.

Q 15 —

$$(7A)^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}$$

As we know that

$$(A^{-1})^{-1} = A$$

Sol) $((7A)^{-1})^{-1} = 7A$

$$\begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}^{-1} = 7A$$

$$\frac{1}{6-7}\begin{bmatrix} -2 & -7 \\ -1 & -3 \end{bmatrix} = 7A$$

$$-1\begin{bmatrix} -2 & -7 \\ -1 & -3 \end{bmatrix} = 7A$$

$$\begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix} = 7A$$

$$A = \frac{1}{7}\begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2/7 & 1 \\ 1/7 & 3/7 \end{bmatrix}$$

Q 16 :—

$$(5A^T)^{-1} = \begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}.$$

As we know that $(A^{-1})^{-1} = A$

$$((5A^T)^{-1})^{-1} = 5A^T$$

$$\frac{1}{-6+5}\begin{bmatrix} 2 & 1 \\ -5 & -3 \end{bmatrix} = 5A^T$$

$$-1\begin{bmatrix} 2 & 1 \\ -5 & -3 \end{bmatrix} = 5A^T$$

$$+\frac{1}{5}\begin{bmatrix} -2 & -1 \\ 5 & 3 \end{bmatrix} = A^T$$

$$A^T = \begin{bmatrix} -2/5 & -1/5 \\ 1 & 3/5 \end{bmatrix}^T$$

$$A^T = \begin{bmatrix} -2/5 & 1 \\ -1/5 & 3/5 \end{bmatrix}$$

Q23:-

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

The matrices A and B commute if $\begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$

$0 = c$ — (1)

$a = d$ — (2)

$0 = 0$ — (3)

$c = 0$ — (4)

Therefore $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ commute if

$$c = 0, \; a = d$$

If we assign b and d the arbitrary values s and t, respectively, the general solution is given by the formulas

$$a = t, \; b = s, \; c = 0, \; d = t.$$

Q24:—

$$AC = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix}$$

The matrices A and C commute if:

$$\begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix} \text{ i.e.}$$

$$b = 0$$
$$0 = 0$$
$$d = a$$
$$0 = b$$

Therefore, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ commute

if $b = 0$, $a = d$

If we assign c and d, the arbitrary values s and t, respectively, the general solution is given by formulas

$a = t$, $b = 0$, $c = s$, $d = t$.

Q 25 :—

Exercise 1.5

Q1:—

(a-) $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$

Q1-4 (On book)

Q5:—

(a-) $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

$EA = \begin{bmatrix} 3 & -6 & -6 & -6 \\ -1 & -2 & 5 & -1 \end{bmatrix}$ ($R_1 \rightarrow R_2$ Interchanged.)

(b-) $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 2 & -1 & 0 & -4 & - \\ 1 & -3 & -1 & 5 & \\ 2 & 0 & 1 & 3 & \end{bmatrix}$

$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ -1 & 9 & 4 & -12 & -10 \end{bmatrix}$ $R_3 - 3R_2$

(c-) $E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

$EA = \begin{bmatrix} -13 & 28 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ $R_1 + 4R_3$

Question 6:—

(a-)

$E = \begin{bmatrix} -6 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

$EA = \begin{bmatrix} 6 & 12 & -30 & 6 \\ 3 & -6 & -6 & -6 \end{bmatrix}$ $\frac{-6 \times R_1}{6}$

(b-)

$E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$

$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ -7 & 1 & -1 & 21 & 19 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$ $R_2 - 4R_1$

(c-)

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$EA = \begin{bmatrix} 1 & 4 \\ 10 & 25 \\ 3 & 6 \end{bmatrix} \quad 5 \times R_2$$

Question 7:—

(a-) $EA = B$

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}$$

B was obtained for interchanging R_1, R_3 of A

So, $$E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(b-) $EB = A$

$$B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

A is obtained from B by interchanging R_1, R_3

$$E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(c-) $EA = C$

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$$

C is obtained for A by ~~obtaining~~ adding $-2R_1$ to R_3

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad (R_3 - 2R_1)$$

(d-) $EC = A$

$$C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 3 \end{bmatrix}$$

A is obtained from C by adding $2R_1$ to R_3

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad (R_3 - 2R_1)$$

Question : 8

(a-) $EB = D$

$$B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix}$$

D is obtained from B by multiplying -3 to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b-) $ED = B$

$$D = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}$$

B is obtained from B by multiplying $-\frac{1}{3}$ to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c-) $EB = F$

$$B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

F is obtained from B by adding $\underline{2R_1 \text{ to } R_2}$.

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

(d-) $EF = B$

$$F = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}$$

B is obtained from F by adding $-2R_1$ to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 9:—

(a-1) $A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix}$

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$ By theorem 1.4.5

$$= \frac{1}{(7)(1)-(4)(2)}\begin{bmatrix} 7 & -4 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -7 & 4 \\ 2 & -1 \end{bmatrix}$$ ✓

By Inversion algorithm

$$A|I = \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right] \quad R_2 - 2R_1$$

$$= \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & 2 & -1 \end{array}\right] \quad -1 \times R_2$$

$$= \left[\begin{array}{cc|cc} 1 & 0 & -7 & 4 \\ 0 & 1 & 2 & -1 \end{array}\right] \quad R_1 - 4R_2$$

$$A^{-1} = \begin{bmatrix} -7 & 4 \\ 2 & -1 \end{bmatrix}$$

(b-)

$$A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -4 \\ -4 & 8 \end{vmatrix}$$

$$= 16 - 16$$
$$= 0$$

The matrix is not invertible.

Question 10:—

a) $$A = \begin{bmatrix} 1 & -5 \\ 3 & -16 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -5 \\ 3 & -16 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -5 \\ 3 & -16 \end{vmatrix}$$

$$= -16 + 15$$

$$= -1$$

$$A^{-1} = \frac{1}{|A|} (\text{Adj } A)$$

$$= \frac{\begin{bmatrix} -16 & 5 \\ -3 & 1 \end{bmatrix}}{-1}$$

$$= \begin{bmatrix} 16 & -5 \\ 3 & -1 \end{bmatrix}$$

By inversion algorithm:—

$$A|I = \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 3 & -16 & 0 & 1 \end{array}\right]$$

$$R_2 - 3R_1 \quad \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right]$$

$$(-1)R_2 \Rightarrow \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right]$$

$$R_1 + 5R_2 \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 16 & -5 \\ 0 & 1 & 3 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 16 & -5 \\ 3 & -1 \end{bmatrix}$$

b) $A = \begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix}$

$$A = \begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 6 & 4 \\ -3 & -2 \end{vmatrix}$$

$$= -12 + 12$$

$$= 0$$

The matrix is not invertible.

Question 11:—

(a-) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

$$[A^{-1}] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - 2R_1$, $R_3 - R_1$ =
$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right]$$

$R_3 + 2R_2$ =
$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & +2 & +1 \end{array}\right]$$

$(-1)R_3$ =
$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$R_1 - 2R_2$ =
$$\left[\begin{array}{ccc|ccc} 1 & 0 & +9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$R_1 - 9R_3$, $R_2 + 3R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

b) $\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

let $A = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

$$|A| = \begin{vmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{vmatrix}$$

$$= +1(-36-2) + 3(-18+4) + 4(4+16)$$

$$= +1(-38) + 3(-14) + 4(20)$$

$$= -38 - 42 + 80$$

$$= 0$$

The matrix is not invertible.

Question : 12

a) $\begin{bmatrix} 1/5 & 1/5 & -2/5 \\ 1/5 & 1/5 & 1/10 \\ 2/5 & -4/5 & 1/10 \end{bmatrix}$

$(5)R_1$, $(5)R_2$, $(5)R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 1 & 1 & 1/2 & 0 & 5 & 0 \\ 1 & -4 & 1/2 & 0 & 0 & 5 \end{array}\right]$$

$R_2 - R_1$, $R_3 - R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \end{array}\right]$$

$\frac{-1}{5} R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 1 & -1/2 & 1 & 0 & -1 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \end{array}\right]$$

$R_1 - R_2$, $\frac{2}{5} R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3/2 & 4 & 0 & 1 \\ 0 & 1 & -1/2 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array}\right]$$

$$\begin{matrix} R_1+\frac{3}{2}R_3 \\ R_2+\frac{1}{2}R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -1 \\ -2 & 2 & 0 \end{bmatrix}$$

(b-)

$$A\,|\,I = \left[\begin{array}{ccc|ccc} 1/5 & 1/5 & -2/5 & 1 & 0 & 0 \\ 1/5 & -3/5 & 3/10 & 0 & 1 & 0 \\ 1/5 & -4/5 & 1/10 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} (5)R_1 \\ (5)R_2 \\ (5)R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 2 & -3 & -3/2 & 0 & 5 & 0 \\ 1 & -4 & 1/2 & 0 & 0 & 5 \end{array}\right]$$

$$\begin{matrix} R_2-2R_3 \\ R_3-R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -10 & -5 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \end{array}\right]$$

$$R_3-R_2 \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -10 & 5 & 0 \\ 0 & 0 & 0 & 5 & -5 & 5 \end{array}\right]$$

Since a row of zero is obtained on left side so A is not invertible.

Question 13:—

$$A = A|I = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$R_3 - R_1 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array}\right]$$

$$R_3 - R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right]$$

$$-\frac{1}{2} R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & 1/2 & -1/2 \end{array}\right]$$

$$\begin{matrix} R_1 - R_3 \\ R_2 - R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 1/2 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1/2 \\ 0 & 0 & 1 & 1/2 & 1/2 & -1/2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}$$

Question 14:—

$$A|I = \left[\begin{array}{ccc|ccc} \sqrt{2} & 3\sqrt{2} & 0 & 1 & 0 & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{\sqrt{2}}\right)R_1$, $\left(\frac{1}{\sqrt{2}}\right)R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ -4 & 1 & 0 & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 + 4R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 13 & 0 & 4/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{13}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 1 & 0 & 2\sqrt{2}/13 & 1/13\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_1 - 3R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \sqrt{2}/26 & -3\sqrt{2}/26 & 0 \\ 0 & 1 & 0 & 2\sqrt{2}/13 & 1/13\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$= \begin{bmatrix} \sqrt{2}/26 & -3\sqrt{2}/26 & 0 \\ 2\sqrt{2}/13 & \frac{1}{13\sqrt{2}} / \frac{\sqrt{2}}{26} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 15:—

$$A = A|I = \left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{array}\right]$$

$$\left(\frac{1}{2}\right)R_1, R_2, R_3 \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 1 & 7/2 & 3 & 0 & 1/2 & 0 \\ 1 & 7/2 & 7/2 & 0 & 0 & 1/2 \end{array}\right]$$

$$\begin{matrix} R_2 - R_1 \\ R_3 - R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & -1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 & -1/2 & 0 & 1/2 \end{array}\right]$$

$$\begin{matrix} 2R_2 \\ 2R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} R_1 - 3R_2 \\ R_3 - R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 7/2 & -3 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$$

$R_1 - 3R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/2 & 0 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$$

$$= \begin{bmatrix} 7/2 & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Question 16:—

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$

Expand by C_4

$$(7) \times \begin{vmatrix} 1 & 3 & 0 \\ 1 & 3 & 5 \\ 1 & 3 & 5 \end{vmatrix}$$

two identical rows or two identical column, their

$|A| = 0$. The matrix is not invertible.

Question 17:—

$$\begin{bmatrix} 2 & -4 & 0 & 0 \\ 1 & 2 & 12 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -4 & -5 \end{bmatrix}$$

let $A = \begin{bmatrix} 2 & -4 & 0 & 0 \\ 1 & 2 & 12 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -4 & -5 \end{bmatrix}$

Expand by C_4

$$= (+5)^{4+4} \begin{vmatrix} 2 & -4 & 0 \\ 1 & 2 & 12 \\ 0 & 0 & 2 \end{vmatrix}$$

Expand by R_3

$$= (+5)(-2)^{3+3}(4+4)$$

$$= -80$$

$$\left[\begin{array}{cccc|cccc} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - 2R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & -8 & -24 & 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{-1}{8}\right)R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 9 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_4 + R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & -5 & -1/8 & 1/4 & 0 & 1 \end{array}\right]$$

$$\left(\frac{1}{2}\right)R_3 \quad \left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/6 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & -1 & -5 & -1/8 & 1/4 & 0 & 1 \end{array}\right]$$

$$R_4+R_3 \quad \left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & -5 & -1/8 & 1/4 & 1/2 & 1 \end{array}\right]$$

$$\left(\frac{1}{-5}\right)R_4 \quad \left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{array}\right]$$

$$R_1-2R_2 \quad \left[\begin{array}{cccc|cccc} 1 & 0 & 6 & 0 & 1/4 & 1/2 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{array}\right]$$

$$\begin{array}{l} R_1-6R_3 \\ R_2-3R_3 \end{array} \quad \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/4 & 1/2 & -3 & 0 \\ 0 & 1 & 0 & 0 & -1/8 & 1/4 & -3/2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/4 & 1/2 & -3 & 0 \\ -1/8 & 1/4 & -3/2 & 0 \\ 0 & 0 & 1/2 & 0 \\ 1/40 & -1/20 & -1/10 & -1/5 \end{bmatrix}$$

Question 18:—

$$A|I = \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & 5 & -3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & 5 & -3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_4 - 2R_2$, $(-1)R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 5 & -5 & 0 & -2 & 0 & 1 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -5 & 0 & -2 & 0 & 1 \end{array}\right]$$

$R_4 - R_2$

$$\left[\begin{array}{cccc|cccc} 2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 8 & -5 & 0 & -2 & 1 & 1 \end{array}\right]$$

$\frac{1}{2} R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 8 & -5 & 0 & -2 & 1 & 1 \end{array}\right]$$

$R_2 + 3R_3$

$R_4 - 8R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & -4 & -2 & 1 & 1 \end{array}\right]$$

$\frac{-1}{5} R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4/5 & 2/5 & -1/5 & -1/5 \end{array}\right]$$

$R_1 - R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -4/5 & 3/5 & 1/5 & 1/5 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4/5 & 2/5 & -1/5 & -1/5 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -4/5 & 3/5 & 1/5 & 1/5 \\ 3/2 & 0 & -1 & 0 \\ 1/2 & 0 & 0 & 0 \\ 4/5 & 2/5 & -1/5 & -1/5 \end{bmatrix}$$

Question 19:—

(a-) $\begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$

let $A = \begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$

$$A^{-1} = \left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{k_1}\right)R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{k}\right)R_1$

$(1/k)R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 1/k & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/k & 0 & 0 & 1/k & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 - \frac{1}{k}R_2 =$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/k & 0 & 0 & 1/k & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{k} - \frac{1}{k}(1)$

$\frac{1}{k}, \frac{-1}{k}, \frac{1}{k}$

$\frac{1}{k}, 0$

$R_3 - \frac{1}{k}R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/k & -1/k \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{cccc} 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/k & -1/k \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Question 20

a)

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

Interchange by C_1 & C_4 & C_2 & C_3

$$\left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & k_2 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_4 & 1 & 0 & 0 & 0 \end{array}\right]$$

$\left(\frac{1}{k_1}\right)R_1$, $\left(\frac{1}{k_2}\right)R_2$, $\left(\frac{1}{k_3}\right)R_3$, $\left(\frac{1}{k_4}\right)R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1/k_1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1/k_2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/k_3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/k_4 & 0 & 0 & 0 \end{array}\right]$$

$$A^{-1} = \left[\begin{array}{cccc} 0 & 0 & 0 & 1/k_1 \\ 0 & 0 & 1/k_2 & 0 \\ 0 & 1/k_3 & 0 & 0 \\ 1/k_4 & 0 & 0 & 0 \end{array}\right]$$

(b-)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ k & 0 & 0 & 0 \\ 0 & 1 & k & 0 \\ 0 & 0 & 1 & k \end{bmatrix}$$

$$\left[\begin{array}{cccc|cccc} k & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & k & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{k}\right)R_1$, $\left(\frac{1}{k}\right)R_2$, $\left(\frac{1}{k}\right)R_3$, $\left(\frac{1}{k}\right)R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 1/k & 1 & 0 & 0 & 0 & 1/k & 0 & 0 \\ 0 & 1/k & 1 & 0 & 0 & 0 & 1/k & 0 \\ 0 & 0 & 1/k & 1 & 0 & 0 & 0 & 1/k \end{array}\right]$$

$R_2 - R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \end{array}\right]$$

$\frac{1}{k}R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 1 & k & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & k & 0 & 0 & -1/k & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{k} R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1/k & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 0 & k & 0 & 0+1/k^2 & 0-1/k & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{k} R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k & 1/k & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{1/k^2}{k} & \frac{-1/k}{k} & 1/k & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{R_4 - R_3}{k}$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 0 & 1 & 0 & 1/k^3 & -1/k^2 & 1/k & 0 \\ 0 & 0 & 0 & 1 & -1/k^4 & 1/k^3 & -1/k^2 & 1/k \end{array}\right]$$

Exercise 1.6

Q1:-

$$x_1 + x_2 = 2$$
$$5x_1 + 6x_2 = 9$$

$$\begin{bmatrix} 1 & 1 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$A \quad X = B$$
$$X = A^{-1}B$$

$$X = \left(\frac{1}{|A|}(Adj\,A)\right) \cdot B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{6-5} \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

So, solution set $= \{3, -1\}$

Q2:—

$$4x_1 - 3x_2 = -3$$
$$2x_1 - 5x_2 = 9$$

$$\begin{bmatrix} 4 & -3 \\ 2 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$AX = B$$
$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\frac{1}{|A|}(\text{Adj } A)\right).B$$

$$= \frac{1}{-20+6}\begin{bmatrix} -5 & 3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$= \frac{1}{-14}\begin{bmatrix} -5 & 3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 5/14 & 3/14 \\ 1/7 & -2/7 \end{bmatrix}\begin{bmatrix} -3 \\ 9 \end{bmatrix}$$

$+\frac{5}{14} - \frac{27}{14}$

$\frac{5\times 9}{14}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \end{bmatrix}$$

$x_1 = -3 \quad , x_2 = -3$

Q3:—

$$x_1 + 3x_2 + x_3 = 4$$
$$2x_1 + 2x_2 + x_3 = -1$$
$$2x_1 + 3x_2 + x_3 = 3$$

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} \\ R_2 - R_3 \\ R_3 - 2R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -1 \\ 0 & -3 & -1 & -2 & 0 & 1 \end{array}\right]$$

$$(-1)\, R_2 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & -3 & -1 & -2 & 0 & 1 \end{array}\right]$$

$$R_3 + 3R_2 = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & -2 & -3 & 4 \end{array}\right]$$

$$(-1)R_3 = \left[\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$R_1 - 3R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & -3 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$R_1 - R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 & 3 & -4 \end{array}\right]$$

$$A^{-1} = \left[\begin{array}{ccc} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & -4 \end{array}\right]$$

$$X = A^{-1} D$$

$$\left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array}\right] = \left[\begin{array}{ccc} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & -4 \end{array}\right] \left[\begin{array}{c} 4 \\ -1 \\ 3 \end{array}\right]$$

$$= \begin{bmatrix} -1 \\ 4 \\ -7 \end{bmatrix}$$

$x_1 = -1$

$x_2 = 4$

$x_3 = -7$

Q4:—

$5x_1 + 3x_2 + 2x_3 = 4$

$3x_1 + 3x_2 + 2x_3 = 2$

$0x_1 + x_2 + x_3 = 5$

$$\begin{bmatrix} 5 & 3 & 2 \\ 3 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 5 & 3 & 2 & 1 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$-(R_1 - 2R_2)$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 3 & 3 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$R_2 - 3R_1 \left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 0 & -6 & -4 & 3 & -5 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left(-\frac{1}{6}\right)R_2 \left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$= R_3 - R_2 \left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 0 & 1/3 & 1/2 & -5/6 & 1 \end{array}\right]$$

$$= (3)R_3 \left[\begin{array}{ccc|ccc} 1 & 3 & 2 & -1 & 2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 0 & 1 & 3/2 & -5/2 & 3 \end{array}\right]$$

$$= R_1 - 3R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 1 & 2/3 & -1/2 & 5/6 & 0 \\ 0 & 0 & 1 & 3/2 & -5/2 & 3 \end{array}\right]$$

$$= R_2 - \frac{2}{3}R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 1 & 0 & -3/2 & 5/2 & -2 \\ 0 & 0 & 1 & 3/2 & -5/2 & 3 \end{array}\right]$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ -3/2 & 5/2 & -2 \\ 3/2 & -5/2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1+0 \\ -6+5-10 \\ 6-5+15 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -11 \\ 16 \end{bmatrix}$$

Q5:—

$$x + y + z = 5 \quad \ldots\ldots$$
$$x + y - 4z = 10$$
$$-4x + y + z = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -4 \\ -4 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1} B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & 1 & 0 \\ -4 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} \\ R_2 - R_1 \\ R_3 + 4R_1 \end{matrix} = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -5 & -1 & 1 & 0 \\ 0 & 5 & 5 & 4 & 0 & 1 \end{array}\right]$$

$$R_2 \leftrightarrow R_3 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 5 & 5 & 4 & 0 & 1 \\ 0 & 0 & -5 & -1 & 1 & 0 \end{array}\right]$$

$$\left(\frac{1}{5}\right)R_2 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & -5 & -1 & 1 & 0 \end{array}\right]$$

$$\left(\frac{-1}{5}\right)R_3 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$R_1 - R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/5 & 0 & -1/5 \\ 0 & 1 & 1 & 4/5 & 0 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$R_2 - R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/5 & 0 & -1/5 \\ 0 & 1 & 0 & 3/5 & 1/5 & 1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 & 0 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/5 & 0 & -1/5 \\ 3/5 & 1/5 & 1/5 \\ 1/5 & -1/5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1/5 & 0 & -1/5 \\ 3/5 & 1/5 & 1/5 \\ 1/5 & -1/5 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 \\ 3+2+0 \\ 1-2+0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -1 \end{bmatrix}$$

Q6:—

$$0w - x - 2y - 3z = 0$$
$$w + x + 4y + 4z = 7$$
$$w + 3x + 7y + 9z = 4$$
$$-w + 2x - 4y - 6z = 6$$

$$\begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 1 & 4 & 4 \\ 1 & 3 & 7 & 9 \\ -1 & -2 & -4 & -6 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \\ 4 \\ 6 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$A^{-1} = \left[\begin{array}{cccc|cccc} 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & 6 & 0 & 0 & 0 & 1 \end{array}\right] \checkmark$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 1 & 3 & 7 & 9 & 0 & 0 & 1 & 0 \\ -1 & -2 & -4 & -6 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_3 - R_1$, $R_4 + R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & -3 & 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array}\right]$$

$-R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 4 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 5 & 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 & 0 & 1 \end{array}\right]$$

$R_3 - 2R_2$, $R_4 + R_2$, $R_1 - R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array}\right]$$

$(-1)R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 & 0 & 1 \end{array}\right]$$

$$
\begin{array}{l}
\\
R_1 - 2R_3 \\
R_2 - 2R_3 \\
R_4 - 2R_3
\end{array}
\left[\begin{array}{cccc|cccc}
1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\
0 & 1 & 0 & 1 & 3 & -2 & 2 & 0 \\
0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\
0 & 0 & 0 & -1 & 3 & -1 & 2 & 1
\end{array}\right]
$$

$$
\begin{array}{l}
\\
\\
\\
-R_4
\end{array}
\left[\begin{array}{cccc|cccc}
1 & 0 & 0 & -1 & 5 & -1 & 2 & 0 \\
0 & 1 & 0 & 1 & 3 & -2 & 0 & 0 \\
0 & 0 & 1 & 1 & -2 & 1 & -1 & 0 \\
0 & 0 & 0 & 1 & -3 & 1 & -2 & -1
\end{array}\right]
$$

$$
\begin{array}{l}
\\
R_2 + R_4 \\
R_2 - R_4 \\
R_3 - R_4
\end{array}
\left[\begin{array}{cccc|cccc}
1 & 0 & 0 & 0 & 2 & 0 & 0 & -1 \\
0 & 1 & 0 & 0 & 6 & -3 & 4 & 1 \\
0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 1 & -3 & 1 & -2 & -1
\end{array}\right]
$$

$$
A^{-1} = \begin{bmatrix}
2 & 0 & 0 & -1 \\
6 & -3 & 4 & 1 \\
1 & 0 & 1 & 1 \\
-3 & 1 & -2 & -1
\end{bmatrix}
$$

$X = A^{-1}B$

$$
\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix}
2 & 0 & 0 & -1 \\
6 & -3 & 4 & 1 \\
1 & 0 & 1 & 1 \\
-3 & 1 & -2 & -1
\end{bmatrix} \begin{bmatrix} 0 \\ 7 \\ 4 \\ 6 \end{bmatrix}
$$

$$
\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -6 \\ 1 \\ 10 \\ -7 \end{bmatrix}
$$

Q7:—

$$3x_1 + 5x_2 = b_1$$
$$x_1 + 2x_2 = b_2$$

$$\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1}B$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{6-5} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{matrix} x_1 \\ x_2 \end{matrix} = \begin{bmatrix} 2b_1 - 5b_2 \\ -b_1 + 3b_2 \end{bmatrix}$$

Q8:—

$$x_1 + 2x_2 + 3x_3 = b_1$$
$$2x_1 + 5x_2 + 5x_3 = b_2$$
$$3x_1 + 5x_2 + 8x_3 = b_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 5 \\ 3 & 5 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$A \quad X = B$$

$$X = A^{-1} B$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = A^{-1} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 5 & 0 & 1 & 0 \\ 3 & 5 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & -1 & 1 & -3 & 0 & 1 \end{array}\right] \quad \begin{matrix} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & -2 & -5 & 1 & 1 \end{array}\right] \quad \begin{matrix} R_1 - 2R_2 \\ R_3 + R_2 \\ \\ \end{matrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 5 & 5 & -2 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5/2 & -1/2 & -1/2 \end{array}\right] \quad -\frac{1}{2}R_3$$

$$
\begin{array}{l}
\\
R_2 + R_3 \\
R_1 - 5R_3
\end{array}
\left[\begin{array}{ccc|ccc}
1 & 0 & 0 & -15/2 & 1/2 & 5/2 \\
0 & 1 & 0 & 1/2 & 1/2 & -1/2 \\
0 & 0 & 1 & 5/2 & -1/2 & 1/2
\end{array}\right]
$$

$$
A^{-1} = \begin{bmatrix} -15/2 & 1/2 & 5/2 \\ 1/2 & 1/2 & -1/2 \\ 5/2 & -1/2 & 2/2 \end{bmatrix}
$$

$$
\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -15/2 & 1/2 & 5/2 \\ 1/2 & 1/2 & -1/2 \\ 5/2 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}
$$

$$
\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{15}{2} b_1 + \frac{1}{2} b_2 + \frac{5}{2} b_3 \\ \frac{1}{2} b_1 + \frac{1}{2} b_2 - \frac{1}{2} b_3 \\ \frac{5}{2} b_1 - \frac{1}{2} b_2 - \frac{1}{2} b_3 \end{bmatrix}
$$

Q9:—

(a-) if $b_1 = 1$, $b_2 = 4$

So, $x_1 - 5x_2 = 1$

$3x_1 + 2x_2 = 4$

In augmented matrix

$$\begin{bmatrix} 1 & -5 & : & 1 \\ 3 & 2 & : & 4 \end{bmatrix}$$

$$R_2 - 3R_1 \begin{bmatrix} 1 & -5 & : & 1 \\ 0 & 17 & : & 1 \end{bmatrix}$$

$$\frac{1}{17} R_2 \begin{bmatrix} 1 & -5 & : & 1 \\ 0 & 1 & : & 1/17 \end{bmatrix}$$

$$R_1 + 5R_2 \begin{bmatrix} 1 & 0 & : & 22/17 \\ 0 & 1 & : & 1/17 \end{bmatrix}$$

So, $x_1 = \frac{22}{17}$ $\quad$ $x_2 = \frac{1}{17}$

(b-) if $b_1 = -2$, $b_2 = 5$

$$x_1 - 5x_2 = -2$$
$$3x_1 + 2x_2 = 5$$

$$\begin{bmatrix} 1 & -5 & : & -2 \\ 3 & 2 & : & 5 \end{bmatrix}$$

$$R_2 - 3R_1 \begin{bmatrix} 1 & -5 & : & -2 \\ 0 & 17 & : & 11 \end{bmatrix}$$

$$\frac{1}{17} R_2 \begin{bmatrix} 1 & -5 & : & -2 \\ 0 & 1 & : & 11/7 \end{bmatrix}$$

$$R_2 + 5R_1 \begin{bmatrix} 1 & 0 & : & 21/17 \\ 0 & 1 & : & 11/17 \end{bmatrix}$$

So, $x_1 = \frac{21}{17}$, $x_2 = \frac{11}{17}$

Q10 :—

$$-x_1 + 4x_2 + x_3 = b_1$$
$$x_1 + 9x_2 - 2x_3 = b_2$$
$$6x_1 + 4x_2 - 8x_3 = b_3$$

$$\begin{bmatrix} -1 & 4 & 1 & : & b_1 \\ 1 & 9 & -2 & : & b_2 \\ 6 & 4 & -8 & : & b_3 \end{bmatrix}$$

(a-) if $b_1 = 0$, $b_2 = 1$, $b_3 = 0$

$$\begin{bmatrix} -1 & 4 & 1 & 0 \\ 1 & 9 & -2 & 1 \\ 6 & 4 & -8 & 0 \end{bmatrix}$$

$$(-1)R_1 \begin{bmatrix} 1 & -4 & -1 & 0 \\ 1 & 9 & -2 & 1 \\ 6 & 4 & -8 & 0 \end{bmatrix}$$

$$\begin{matrix} R_2 - R_1 \\ R_3 - 6R_1 \end{matrix} \begin{bmatrix} 1 & -4 & -1 & 0 \\ 0 & 13 & -1 & 1 \\ 0 & 28 & -2 & 0 \end{bmatrix}$$

$$\frac{1}{13} R_2 \begin{bmatrix} 1 & -4 & -1 & 0 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 28 & -2 & 0 \end{bmatrix}$$

$$\begin{matrix} R_1 + 4R_2 \\ R_3 - 28R_2 \end{matrix} \begin{bmatrix} 1 & 0 & -17/13 & 4/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 2/13 & -20/13 \end{bmatrix}$$

$$\frac{13}{2} R_3 \begin{bmatrix} 1 & 0 & -17/13 & 4/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -14 \end{bmatrix}$$

$$\begin{matrix} R_1 + \frac{17}{13} R_3 \\ R_2 + \frac{1}{13} R_3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & -18 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -14 \end{bmatrix}$$

$x_1 = -18$, $x_2 = -1$, $x_3 = -14$

(b-) if $b_1 = -3$, $b_2 = 4$, $b_3 = -5$

$$\begin{bmatrix} -1 & 4 & 1 & -3 \\ 1 & 9 & -2 & 4 \\ 6 & 4 & -8 & -5 \end{bmatrix}$$

$$(-1)R_1 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 1 & 9 & -2 & 4 \\ 6 & 4 & -8 & -5 \end{bmatrix}$$

$$\begin{matrix} \\ R_2 - R_1 \\ R_3 - 6R_1 \end{matrix} \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 13 & -1 & 1 \\ 0 & 28 & -2 & -23 \end{bmatrix}$$

$$\frac{1}{13}R_2 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 28 & -2 & -23 \end{bmatrix}$$

$$R_3 - 28R_2 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 2/13 & 327/13 \end{bmatrix}$$

$$\frac{13}{2}R_3 \begin{bmatrix} 1 & -4 & -1 & 3 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & 327/2 \end{bmatrix}$$

$$R_1 + 4R_2 \begin{bmatrix} 1 & 0 & -17/13 & 43/13 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -327/2 \end{bmatrix}$$

$$R_1 + \frac{17}{13}R_3 \begin{bmatrix} 1 & 0 & 0 & 5645/26 \\ 0 & 1 & -1/13 & 1/13 \\ 0 & 0 & 1 & -327/2 \end{bmatrix}$$

$$\frac{43}{13}\ R_2 + \frac{1}{13}R_3 \begin{bmatrix} 1 & 0 & 0 & 5645/26 \\ 0 & 1 & 0 & \\ 0 & 0 & 1 & 327/2 \end{bmatrix}$$

SB (Complete it yourself).

$x_1 = \frac{-421}{2}$ $x_2 = \frac{-25}{2}$ $x_3 = \frac{-327}{2}$

Q 11:—

$$4x_1 - 7x_2 = b_1$$
$$x_1 + 2x_2 = b_2$$

$$\begin{bmatrix} 4 & -7 & b_1 \\ 1 & 2 & b_2 \end{bmatrix}$$

$$\left[\begin{array}{cc:c:c:c:c} 4 & -7 & 0 & -4 & -1 & -5 \\ 1 & 2 & 1 & 6 & 3 & 1 \end{array}\right]$$

(a-) (b-) (c-) (d-)

$R_1 \leftrightarrow R_2$)
$$\left[\begin{array}{cc:c:c:c:c} 1 & 2 & 1 & 6 & 3 & 1 \\ 4 & -7 & 0 & -4 & -1 & -5 \end{array}\right]$$

$R_2 - 4R_1$
$$\left[\begin{array}{cc:c:c:c:c} 1 & 2 & 1 & 6 & 3 & 1 \\ 0 & -15 & -4 & 28 & -13 & -9 \end{array}\right]$$

$\frac{-1}{15} R_2$
$$\left[\begin{array}{cc:c:c:c:c} 1 & 2 & 1 & 6 & 3 & 1 \\ 0 & 1 & 4/15 & 28/15 & 13/15 & 3/5 \end{array}\right]$$

$$R_1 - 2R_2 \begin{bmatrix} 1 & 0 & : 7/15 & : 34/15 & : 19/15 & : -1/5 \\ 0 & 1 & : 4/15 & : 28/15 & : 13/15 & : 3/5 \end{bmatrix}$$

(a-) $x_1 = 7/15$, $x_2 = 4/15$

(b-) $x_1 = 34/15$, $x_2 = 28/15$

(C-) $x_1 = 19/15$, $x_2 = 13/15$

(d-) $x_1 = -1/5$, $x_2 = 3/5$

Q12 :—

$$\begin{bmatrix} 1 & 3 & 5 & : b_1 \\ -1 & -2 & 0 & : b_2 \\ 2 & 5 & 4 & : b_3 \end{bmatrix}$$

$$\begin{bmatrix} +1 & 3 & 5 & : 1 & : 0 & : -1 \\ -1 & -2 & 0 & : 0 & : 1 & : -1 \\ 2 & 5 & 4 & : -1 & : 1 & : 0 \end{bmatrix}$$

(a-) (b-) (c-)

$$\begin{matrix} \\ R_2 + R_1 \\ R_3 - 2R_1 \end{matrix} \begin{bmatrix} 1 & 3 & 5 & : 1 & : 0 & : -1 \\ 0 & 1 & 5 & : 1 & : 1 & : -2 \\ 0 & -1 & -6 & : -3 & : 1 & : 2 \end{bmatrix}$$

$$\begin{matrix} \\ R_1-3R_2 \\ R_3+R_2 \end{matrix}\begin{bmatrix} 1 & 0 & -10 & : & -2 & : & -3 & : & 5 \\ 0 & 1 & 5 & : & 1 & : & 1 & : & -2 \\ 0 & 0 & -1 & : & -2 & : & 2 & : & 0 \end{bmatrix}$$

$$(-1)R_3\begin{bmatrix} 1 & 0 & -10 & : & -2 & : & -3 & : & 5 \\ 0 & 1 & 5 & : & 1 & : & 1 & : & -2 \\ 0 & 0 & 1 & : & 2 & : & -2 & : & 0 \end{bmatrix}$$

$$\begin{matrix} R_2-5R_3 \\ R_1+10R_3 \end{matrix}\begin{bmatrix} 1 & 0 & 0 & : & 18 & : & -23 & : & 5 \\ 0 & 1 & 0 & : & -9 & : & 11 & : & -2 \\ 0 & 0 & 1 & : & 2 & : & -2 & : & 0 \end{bmatrix}$$

(a-) $x_1 = 18$, $x_2 = -9$, $x_3 = 2$

(b-) $x_1 = -23$, $x_2 = 11$, $x_3 = -2$

(c-) $x_1 = 5$, $x_2 = -2$, $x_3 = 0$

Q13:—

$$\begin{bmatrix} 1 & 3 & b_1 \\ -2 & 1 & b_2 \end{bmatrix}$$

$$R_2+2R_1\begin{bmatrix} 1 & 3 & b_1 \\ 0 & 7 & b_2+2b_1 \end{bmatrix}$$

$$\frac{1}{7}R_2\begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & \frac{b_2+2b_1}{7} \end{bmatrix}$$

$$R_1 - 3R_2 \begin{bmatrix} 1 & 0 & \frac{7b_1 - 3b_2 + 6b_1}{7} \\ 0 & 1 & \frac{b_2 + 2b_1}{7} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & \frac{13b_1 - 3b_2}{7} \\ 0 & 1 & \frac{b_2 + 2b_1}{7} \end{bmatrix}$$

System has unique solution

$$x_1 = \frac{13b_1 - 3b_2}{7}$$

$$x_2 = \frac{b_2 + 2b_1}{7}$$

Q14:—

$$6x_1 - 4x_2 = b_1$$
$$3x_1 - 2x_2 = b_2$$

$$\begin{bmatrix} 6 & -4 & b_1 \\ 3 & -2 & b_2 \end{bmatrix}$$

$$\frac{1}{6}R_1 \begin{bmatrix} 1 & -2/3 & b_1/6 \\ 3 & -2 & b_2 \end{bmatrix}$$

$$R_2 - 3R_1 \begin{bmatrix} 1 & -2/3 & b_1/6 \\ 0 & 0 & b_2 - \frac{b_1}{2} \end{bmatrix}$$

For system to be consistent

$$b_2 - \frac{b_1}{2} = 0$$

$$b_2 = \frac{b_1}{2} \qquad b_1 = 2b_2$$

Q15:—

$$\begin{bmatrix} 1 & -2 & 5 & b_1 \\ 4 & -5 & 8 & b_2 \\ -3 & 3 & -3 & b_3 \end{bmatrix}$$

$$\begin{matrix} R_2 - 4R_1 \\ R_3 + 3R_1 \end{matrix} \begin{bmatrix} 1 & -2 & 5 & b_1 \\ 0 & 3 & -12 & b_2 - 4b_1 \\ 0 & -3 & 12 & b_3 + 3b_1 \end{bmatrix}$$

$$R_3 + R_2 \begin{bmatrix} 1 & -2 & 5 & b_1 \\ 0 & 3 & -12 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 + b_2 - b_1 \end{bmatrix}$$

system is consistent because $R_3 = 0$.

$$b_3 + b_2 - b_1 = 0 \Rightarrow b_1 = b_3 + b_2$$

Q16 :—

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ -4 & 5 & 2 & b_2 \\ -4 & 7 & 4 & b_3 \end{array}\right]$$

R_2+4R_1, R_3+4R_1

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -3 & -2 & b_2+4b_1 \\ 0 & -1 & 0 & b_3+4b_2 \end{array}\right]$$

$\frac{-1}{3}R_2$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -\frac{(b_2+4b_1)}{3} \\ 0 & -1 & 0 & b_3+4b_1 \end{array}\right]$$

R_3+R_2

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -\frac{b_2-4b_1}{3} \\ 0 & 0 & 2/3 & \frac{3b_3-b_2+8b_1}{3} \end{array}\right]$$

$\frac{3}{2}R_3$, R_2-R_3

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 2/3 & -4b_1-b_3 \\ 0 & 0 & 2/3 & \frac{3b_3-b_2+8b_1}{3} \end{array}\right]$$

$\frac{3}{2}R_3$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{array}\right]$$

R_1+2R_2

$$\begin{bmatrix} 1 & 0 & -1 & -7b_1-2b_3 \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{bmatrix}$$

R_3+R_3

$$\begin{bmatrix} 1 & 0 & 0 & \frac{-6b_1-b_2-b_3}{2} \\ 0 & 1 & 0 & -4b_1-b_3 \\ 0 & 0 & 1 & \frac{3b_3-b_2+8b_1}{2} \end{bmatrix}$$

$$x_1 = \frac{-6b_1-b_2-b_3}{2}$$

$$x_2 = -4b_1-b_3$$

$$x_3 = \frac{3b_3-b_2+8b_1}{2}$$

Q17:—

$$\begin{bmatrix} 1 & -1 & 3 & 2 & b_1 \\ -2 & 1 & 5 & 1 & b_2 \\ -3 & 2 & 2 & -1 & b_3 \\ 4 & -3 & 1 & 3 & b_4 \end{bmatrix}$$

	1	-1	3	2	b_1
R_2+2R_1	0	-1	11	5	b_2+2b_1
R_3+3R_2	0	-1	11	5	b_3+3b_1
R_4-4R_1	0	1	-11	-5	b_4-4b_1

R_4+R_2	1	-1	3	2	b_1
R_3-R_2	0	-1	11	5	b_2+2b_1
	0	0	0	0	$b_3+b_1-b_2$
	0	0	0	0	$b_4+b_2-2b_1$

So the system is consistant if

$$b_4+b_2-2b_1=0$$

$$b_1=-b_2+b_3$$

$$2$$